

AS • Edexcel • Physics

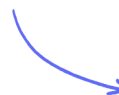
 3 mins 3 questions

Multiple Choice Questions

# The Photoelectric Effect & Atomic Spectra

The Photoelectric Effect / The Photoelectric Equation / The Electronvolt / The Particle Nature of EM Radiation / Atomic Line Spectra

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Total Marks

/3

- 1 The photoelectric effect provides evidence for the particle nature of electromagnetic radiation.

Which of the following observations of the photoelectric effect could also be explained using the wave nature of electromagnetic radiation?

- A. The emission of photoelectrons is instantaneous.
- B. The maximum kinetic energy of photoelectrons depends on frequency.
- C. The rate of emission of photoelectrons depends on intensity.
- D. There is a minimum frequency for emission of photoelectrons to occur.

**(1 mark)**

- 2 When monochromatic light is incident on the surface of a metal, electrons are emitted by the photoelectric effect.

If other conditions are unchanged, the maximum kinetic energy of the electrons will be increased by

- A. increasing the frequency of the incident light.
- B. increasing the intensity of the incident light.
- C. using a metal with a higher threshold frequency.
- D. using a metal with a higher work function.

**(1 mark)**

- 3** When light is incident on the surface of a metal, electrons may be emitted by the photoelectric effect. Observations of the photoelectric effect helped to establish that light can exhibit particle behaviour.

Which of the following observations of the photoelectric effect could also be explained by light behaving as a wave?

- A.** Emission of photoelectrons occurs immediately the surface is illuminated.
- B.** Photoelectrons are only emitted when the frequency of the light is more than a certain minimum value.
- C.** The maximum kinetic energy of the photoelectrons is independent of the intensity of the incident light.
- D.** When the intensity of the incident light increases, photoelectrons are emitted at a greater rate.

**(1 mark)**